UEscope®

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UEscope®

Video Laryngoscopes

Product manual for operation, maintenance
and technical instructions
(November, 2016)
B/0

Zhejiang UE Medical Corp.

Thank you for purchasing our products.

⚠ This product can be used only by medical practitioners and medical professionals going through operation training.

⚠ Please read the manual carefully before using the product and keep properly for future use.

Please abide by the operation procedure strictly for correct maintenance when using this product.

⚠ Paragraphs marked with "Note" shall be read and implemented carefully to avoid injury or damage to devices, operators and patients.

If there is any problem with devices during use, please contact the local dealer or the Company, and we will provide you with quality services and assistance.

Commitment: We are committed to provide more detailed necessary technical data to users when processing failures if you need.

⚠ Since the product is improved continuously, the appearance or specification of the product provided by the Company may be different from that stated in this instruction, which is hereby informed in advance.

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Appendix EMC Performance Index

⚠ Warning:

1. This product cannot be used in places where there are fire hazards.

2. This product cannot be used in situations where there is a mixture of inflammable anesthetizing gas and air or nitrous oxide.

3. Inside this product, there are special devices, and users shall not disassemble or refit this product by themselves.

4. This product shall not be used while charging.

5. The TDC and TD-C laryngoscope can only be used together with laryngoscope blade for single use with registration certificate.

6. The blade of TDC and TD-C laryngoscope can not be disinfected by soaking in the liquid.

7. This product applies cold light for surgery, please reserve one Video Laryngoscopes.

8. The light on the front end of the laryngoscope cannot strictly shine into patients' or other persons' eyes.

⚠ Safety Advice :

1. This product must be protected from any external adverse effects such as strong electromagnetic radiation or high temperature.

2. During transportation and use, this product shall be handled with care, and prevented from impact, violent vibration and humidity.

3. This product can be repaired only by professionals authorized by the Company.

4. This product can be charged only by using the charger provided by the Company.

5. In case of any failure in this product, please contact the customer service department of our company in time.

⚠ Notes:

1. Do not overexert yourself in use.

2. The display component of VL laryngoscope and the whole machine of TDC and TD-C laryngoscope are not waterproof. Please pay attention to water resistance when cleaning.

3. Check this product before use and make sure there is no rust, indentation, no scratch, sharp edge or protrusion.

4. For the reusable Video Laryngoscopes, please disinfect it with glutaraldehyde disinfectant before use.

5. Please charge the product for the first use.

6. When out of service for a long time, charge this product every 2-3 months.

Contraindication: None.

Part 1

1.1 Application Scope:

The Video Laryngoscopes (hereinafter referred to as the laryngoscope) is used for leading in tracheal intubation in anesthetization and emergency treatment of medical institutions.

1.2 Main Structure:

VL laryngoscope is composed of display component, blade component (camera, LED light) and charger.See the diagram below.

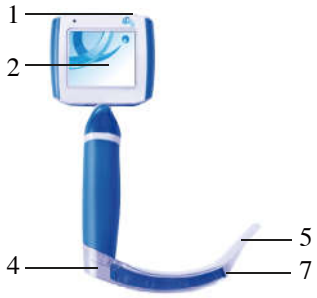
TD-C and TDC laryngoscope are composed of display component, blade component (camera, LED light) and charger.See the diagram below.

Note:
VL laryngoscope includes VL300 laryngoscope and VL310 laryngoscope; TDC laryngoscope includes TDC-C laryngoscope and TDC-K laryngoscope.

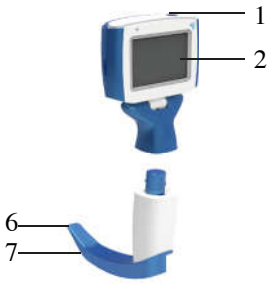
TDC-C

TDC-K

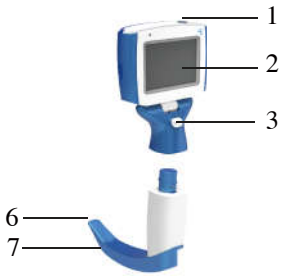
1. Switch Button
2.LCD Screen
3. Function Key
4. Blade Holder
5. Laryngoscope Blade for Single Use
6. Blade
7. Camera



TD-C



VL300



VL310

Series Difference Table (Table 1.1)

VL300 Series	Split, non-storage, use directly
VL310 Series	Split, storage, use directly
TD-C Series	One-piece, non-storage, use with laryngoscope blade for single use
TDC-C Series	Split, non-storage,use with laryngoscope blade for single use
TDC-K Series	Split, storage,use with laryngoscope blade for single use

The internal cantilever double-clip rotation structure design with unique patented technologies is adopted for the VL and TDC laryngoscope, which is connected simply, safely and reliably, avoiding poor connection phenomenon due to handle looseness; the sharp angle and radian of the VL laryngoscope blade and laryngoscope blade for single use with unique patented technologies maximizes the focus definition of the camera, and makes tracheal intubation much easier; the tangent design is adopted for the field edge and laryngoscope sharp of the camera reduces the dead zone of the cannula field greatly.

The horizontal screen design of the laryngoscope expands the working vision effectively, presents the horizontal position of the laryngoscope blade and tracheal intubation perfectly, reduces the image morphing and shortens the height of the complete machine.

1.3 Model

Table of Models (Table 1.2)

Series	Model				
VL300 Series	VL300L	VL300M	VL300SM	VL300S	VL300SS-1
Vl310 Series	VL310-4	VL310-3	VL310-2	VL310-1	VL310-0
TDC-C Series	TDC-C4	TDC-C3	TDC-C2	TDC-C1	TDC-C0
TDC-K Series	TDC-K4	TDC-K3	TDC-K2	TDC-K1	TDC-K0
TD-C Series	TD-C-V	TD-C-IV	TD-C-III	TD-C-II	TD-C-I

The TDC and TD-C laryngoscope shall be used in conjunction with the laryngoscope blade for single use with registration certificate, and the laryngoscope shall not contact the human body directly. The VL laryngoscope can be used repeatedly, and the laryngoscope can contact the human body directly.

Corresponding relationship between TDC and TD-C laryngoscope and laryngoscope for single use (Table 1.3)

TDC-C Series	TDC-K Series	TD-C Series	Laryngoscope Blade for Single Use
TDC-C0	TDC-K0	TD-C-I	TDC-0
TDC-C1	TDC-K1	TD-C-II	TDC-1
TDC-C2	TDC-K2	TD-C-III	TDC-2
TDC-C3	TDC-K3	TD-C-IV	TDC-3
TDC-C4	TDC-K4	TD-C-V	TDC-4

Part 2

2.1 Basic Performance Indicators

Category of the device: Internal power supply.

Type of the device: BF type device with intermittent operation

Applied part of the device: blade, laryngoscope blade for single use

The display component of the device is a common non-watertight device.

The device is a portable device (Table 1.4)

Name	Technical Name	Technical Specifications
Parameters of the complete machine	Resolution ratio	⩾3.72 lp/mm
	Angle of vision	$60^{0}\pm15\%$
	Standby time	>2h
	Charging times	＞300 times
	Display	2.5″ TFT
	Power consumption	⩽2VA
	Illuminance	⩾150LUX
	Front-and-rear rotation angle of the display	$0^{0}\sim130^{0}$
	Left-and-right rotation angle of the display	$0^{0}\sim270^{0}$
	Maximum storage space (only applicable to TDC-K and VL310 series)	32G
	Maximum photo storage space (only applicable to TDC-K and VL 310 series)	100,000 photos
	Maximum video time (only applicable to TDC-K and VL 310 series)	4.5 hours

Name	Technical Name	Technical Specifications
Charger (Class II)	Charger input	100~240V，50-60Hz
	Charger output	5VDC，1.2A
	Charging time	<3 hours
Working environment	Temperature	-5～+50℃
	Humidity	10%～90%
	Atmospheric pressure	860hPa～1060hPa
Transportation and storage environment	Temperature	-40℃～+55℃
	Humidity	≤93%
	Atmospheric pressure	500hPa～1060hPa

2.2 Instructions for Use

Operating switch:

Press the switch button for 1 to 3 seconds at startup and release it after startup. Press the switch button for 1 to 3 seconds at shutdown and release it after shutdown.

Camera:

Short press the camera button, and the photo symbol on the top left corner of the screen will flash, meaning taking a photo, the format of the photo is JPG, and the pixel of the photo is 640*480. (Only applicable to TDC-K and VL310 series products.)

Real-time video:

Long press the camera button for 1 second, and the flickering red dot will appear on the top left corner of the screen and timing will start on the top right corner of the screen, meaning it is videoing. Long press the camera button for 1 more second and exit video. The format of the video is AVI, and the pixel of the photo is 640*480. (Only applicable to TDC-K, VL310 series products.)

Data transmission:

Use a matched data line with one end connecting the charging interface of the display and the other end connecting USB interface of the computer, and long press the startup button to start up the machine. After normal connection with the computer, the display area of the laryngoscope will darken automatically. If the computer is connected to the laryngoscope for the first time, an USB drive will be installed automatically. After connection, open the “removable disk” in the computer, and data access operation can be conducted.

The instrument with 8G memory can take at least 100,000 photos or video for 4.5 hours. After the TF card memory is full, it cannot be stored circularly, and a computer shall be connected to delete manually. (Only applicable to TDC-K, VL310 series products.)

Operating state:

The green light is on in normal operation state, the red light is on in low battery, and please recharge in time. Meanwhile, the battery symbol on the top left corner of the screen will indicate the electric quantity in grade division.

Charging:

Please shut down while it is charging, and use the standard AC 220V 50 Hz power source for the charger. The green indicator light of the display will flicker while it is charging, and the green light will be normally on after a full charge.

2.3 Instructions for Operation

1. ***VL and TDC laryngoscope:*** Hold connectors still in one hand and the handle in the other hand, insert the laryngoscope blade component into the display component according to the specific position, rotate in place counterclockwise, and be careful not to grip hands by connectors and the handle when inserting. As for the TDC and TD-C laryngoscope, laryngoscope blade for single use shall be inserted into the holder pedestal component of the Video Laryngoscopes according to specific positions, and clips of the laryngoscope blade for single use shall be clipped in place. Turn on the power, and the power indicator will be on, and the display will show the scene captured by the high definition camera.

2.The operation method of Video Laryngoscopes is basically the same as that of common direct laryngoscope. The patient shall be in supine position, the operator shall stand in head orientation of the patient, pull the patient' s upper incisors with the right index finger to make the atlanto-occipital joint stretch, push the region of chin downward with the right middle finger to make the patient open mouth, and hold the Video Laryngoscopes with the left hand to insert the laryngoscope blade in the patient' s mouth in the tongue center. Make the laryngoscope blade bend in normal direction of the oral cavity and pharyngeal region, slide into the pharyngeal region downward slowly on the surface of the tongue, and the root of tongue, palate and epiglottis can be seen successively in the display.

3. Place the front end of the laryngoscope blade on the epiglottic vallecula and lift the Video Laryngoscopes gently to expose the glottis in the display; if the glottis is exposed poorly, it is suggested to adopt the mandibular lifting operation for assistance.

4. After exposing the glottis clearly, insert the tracheal tube with intubating stylet and 60° of the front end moulding into the patient' s mouth from the right side of the laryngoscope blade; once the front end of the tracheal tube enters the front view of the laryngoscope blade, the relation between the front end of the tracheal tube and the glottis will be clearly showed in the display; align the front end of the tracheal tube with the glottis and enter the infraglottic portion slightly; while the assistant pulls out the intubating stylet, the operator shall continue to push the tracheal tube downward under the monitoring of the display.

5. Insert the tracheal tube to an appropriate depth (about 1 cm subglottic the cuff), and the operator shall fix the tracheal tube with the right hand and remove the laryngoscope blade from the mouth with the left hand. Check the relation between the scale mark of the surface of the tracheal tube and the incisor, so as to further affirm the inserting depth of the tracheal tube.
Then inflate the cuff of the tracheal tube appropriately, and connect to a ventilator to control respiration.

6. After use of VL and TDC laryngoscope, hold the connector still in one hand and the handle in the other to unscrew and pull out the laryngoscope, and place the display component in the box. As for the VL laryngoscope, the laryngoscope component shall be disinfected before being placed into the box. As for the TDC laryngoscope, pull out the laryngoscope blade for single use from the laryngoscope blade holder base, and abandon the laryngoscope blade for single use for recycling.

Part 3

3.1 Influence Factors on the Use:

1.Not reading the operation instruction carefully or lack of operation experience.

2.Excessive low environmental temperature and intensive sun exposure in the outdoor environment.

3.Curvature of the front end of the tracheal tube with inbubating stylet.

4.Unsatisfying lubrication of the laryngoscope blade or laryngoscope blade for single use.

5.Unreasonable insertion and position of the laryngoscope blade or laryngoscope blade for single use.

3.2 Clinical Application Notes:

The operation technique of Video Laryngoscopes is basically the same as that of common direct laryngoscope. Therefore, all medical staff who can operate direct laryngoscope proficiently, after read the instructions of use carefully can operate Video Laryngoscopes freely without special training. However, the following factors influencing the application of Video Laryngoscopes to conduct the tracheal tube operation shall be paid special attention to:

1. If the environmental temperature is too low, do not insert the Video Laryngoscopes into the patient’s mouth immediately after turning on the switch, and wait for 1 minute to preheat the camera and protection cover at the front end of the laryngoscope blade, to prevent the blade from being covered with vapour in tracheal intubation, making it unable to identify the pharyngolaryneal structure.

2. The strong sunshine can reduce the display of the airway structure and tracheal intubation operation in the liquid crystal display, and it is not recommended to apply it in outdoor environment with strong sunshine.

3. The intubating stylet with a certain elasticity and hardness shall be selected, and an angle of about 60° shall be formed at the front end of the tracheal intubation with intubating stylet, to guarantee that the front end of the tracheal intubation can directly face the glottis and the tracheal intubation operation can be implemented successively.

4. Though appropriate amount of lubricant is applied to lubricate the ventral side of the laryngoscope blade before use every time, it is not suitable to apply too much lubricant so as to avoid excessive lubrication and make it difficult to fix the laryngoscope blade in the mouth.

5. If vapour is created on the protection cover at the front end of the laryngoscope blade due to the patient’s coughing and autonomous respiration, or lots of blood or secreta in the mouth contaminates the front end of the laryngoscope blade, the laryngoscope shall be removed from the mouth, and vapour or dirt shall be removed with a soft gauze or fine tissue before the tracheal intubation operation.

6. The laryngoscope blade is inserted too deep. If the laryngoscope blade is inserted into the throat too deep, the whole laryngeal may be lifted, so what is revealed is the gullet opening rather than the glottis opening. In the course of tracheal intubation, insert the Video Laryngoscopes slowly along the center line of the oral cavity, and the root of tongue, palate and epiglottis can be seen successively, avoiding the occurrence of such problem effectively.

7. As for women with emphysematous chest, obesity and big breasts and patients suffering from chin-thorax scar conglutination, it may be difficult to insert the Video Laryngoscopes into the mouth as the patients’ anterior chest wall may impede the movement of the laryngoscope handle and display. Larger operation space may be obtained for the Video Laryngoscopes by further stretching the atlanto-occipital joint and leaning back the patient's head. Besides, by rotating the Video Laryngoscopes for 90 degree to the right, the laryngoscope can often be inserted into the patient's mouth smoothly, and interference of the anterior chest wall with the Video Laryngoscopes operation can be avoided.

8. If it is difficult to push the tracheal tube downward after removing the intubating stylet, the strength of lifting the laryngoscope may be reduced to return the patient’s head to the neutral position state, and the tracheal tube shall be pushed downward.

3.3 Maintenance, Cleaning and Disinfection:

1.Disinfection of the laryngoscope blade in VL product family: Put the laryngoscope blade into the container filled with 0.5%~0.6% ortho-phthalaldehyde disinfectant and soak it for 20 minutes, wash it with normal saline, and put it in the aseptic bag for further use.

2.Disinfection of the laryngoscope blade in VL product family: Put the laryngoscope blade into the container filled with 2% glutaraldehyde disinfectant and soak it for 20 minutes, wash it with normal saline, and put it in the aseptic bag for further use.

3.When cleaning the laryngoscope blade, do not use the hard matter to scratch the camera protection cover at the front end of the laryngoscope blade to prevent impacting the image definition.

4.Routine maintenance of TDC and TD-C laryngoscope: Use a slightly wet cotton ball to wipe the surface after use.

⚠ The display of VL laryngoscope and the whole machine of TDC and TD-C laryngoscope are not waterproof, and it is forbidden to soak it for disinfection.

⚠ It is not recommended to use strong acid and strong oxidizing disinfectant, or adopt other disinfection methods beyond the product's working environmental requirements.

Part 4

4.1 Storage and Transportation Conditions

During the transportation process, keep the cleanliness without any pollution, and it is forbidden to be soaked or drenched by rain and snow.

Store it inside the room without harmful gas and infringement of foreign bionts and with good ventilation.

4.2 Troubleshooting

Fault	Reasons	Troubleshooting
Unclear	Lens was Polluted	Cleaning Lens
No Starting up	No Power in Batteries	Charging
No Image after Starting up	Bad Contact of the Laryngoscope Blade and Display Component	Cleaning the Contact Point of the Laryngoscope Blade and Display Component and Starting up again

4.3 Product Recovery

Treatment of the Laryngoscope and Accessories during the End Period of Service Life: Belong to the nondegradable articles.

Advice: Hand over the laryngoscope and accessories during the end period of service life to the environment protection department for the recovery.

4.4 Warranty Period

There is 1-year maintenance for the complete machine from the selling date of the product.

⚠ **Note:** For the infection management and safety of personnel maintaining the machines, before any machine is delivered to Zhejiang UE for maintenance, the machines shall be completely washed and disinfected by high-level disinfection operation. If the machine is provided for the use of patients with the HA positive or other infectious disease, please inform the manufacturer.

EC REP 4.5 Information of European Agency

European Authorized Representative: Lotus Global Co., Ltd
Address: 1 Four Seasons Terrance West Drayton , Middlesex London, UB79GG United Kingdom

4.6 Production Date

Production Date :See the product label

4.7 Customer Service

Zhejiang UE Medical Corp.

Address: No. 8, Youyi Road, Baita Economic Development Zone, Xianju , Zhejiang317300 , China

Tel: + 86 576 87788798 Fax : + 86 576 87721129

E-mail: info@ueworld.com

Website: www.ueworld.com

4.8 Mark Index(Table1.5)

Mark	Meaning
	BF-Type Equipment
	Switch Button
	Photographing
	Note, Reference to the Attached Documents
	Reference to the Instructions for Use
	Indication of the Electric Quantity
	Low Electric Quantity
	Direct Current
	Manufacturer
	Production Date
EC REP	Information of European Agency
SN	Product Series No.
	Separate handling of the waste electrical and electronic equipment(Please comply with local laws and regulations)

Appendixes: EMC Performance Index

Basic Performance of the Product (Table 1.6)

Names	Concrete Description
Working Mode	Under any situations, the liquid crystal display can conduct the real-time display of camera, and the display can display the camera images at the same time.
Charging Mode	Under the situation that the battery capacity is not full, the indication light will flicker, and after the battery capacity is full, the indication light will be green all the time.
Data Transmission Mode(Only applicable for the series of TDC-K and VL310)	Successfully read the data.

The special precautionary measures related to the electromagnetic compatibility (EMC) shall be adopted for the device, and the installation and use shall be conducted in accordance with the information on the electromagnetic compatibility stipulated in the instructions. Portable and mobile radio-frequency communication equipment may affect the device.

The following cable shall be used to meet the requirements on the aspect of the electromagnetic emission and anti-interference performance. (Table 1.7)

Cable Name	Length
USB Date Line	1.4 meters

In addition to the cables (transducers) sold as the spare parts of the internal components, the accessories and cables (transducers) which are not stipulated may increase the emission of the device or system or reduce the noise immunity.

The device or system shall not be near to or stacked up with other equipment for the use, and if the device or system needs to be near to or stacked up with other equipment, the observation and verification shall be conducted for ensuring that the normal operation can be achieved under its used configuration.

Guideline and Statement of the Manufacturer – Electromagnetic Emission (Table 1.8)

Guideline and Statement of the Manufacturer – Electromagnetic Emission		
The Video Laryngoscopes is estimated to be used in the following stipulated electromagnetic environment, and the purchasers or users shall guarantee that it is used in the electromagnetic environment.		
Emission Experiment	Conformity	Electromagnetic Environment – Guideline
Radio-frequency Emission GB4824	1 Group	The Video Laryngoscopes only uses the radio-frequency energy for the internal functions. Therefore, its radio-frequency emission is very low, and has very low possibility for producing the disturbance for the nearby electric equipment.
Radio-frequency Emission GB4824	Classification A	The Video Laryngoscopes is applicable to the use of all facilities of non-domestic and domestic residence not directly connected with the public low-voltage power supply network.
Harmonic Emission GB 17625.1	Not Applicable	
Voltage Fluctuation /Flicker Emission GB 17625.2	Not Applicable	

Guideline and Statement of the Manufacturer – Electromagnetic Immunity (Table 1.9)

Guideline and Statement of the Manufacturer – Electromagnetic Immunity			
The Video Laryngoscopes is estimated to be used in the following stipulated electromagnetic environment, and the purchasers or users shall guarantee that it is used in the electromagnetic environment.			
Immunity Test	IEC 60601 Test Level	Conformity Level	Electromagnetic Environment – Guideline
Electrostatic Discharge GB/T 17626.2	±6 kV Contact Discharge ±8 kV Air Discharge	±6 kV Contact Discharge ±8 kV Air Discharge	The ground shall be wood, concrete or tile, and if the ground is covered by the synthetic material, the relative humidity shall be 30% at least.
Electrical Fast Transient Burst GB/T 17626.4	±2kV for Power Supply Line ±1kV for Input/Output Line	±2kV for Power Supply Line Not Applicable	The network power supply shall be equipped with the quality used in the classic commercial or medical environment.
Surge GB/T 17626.5	±1kV Wire to Wire ±2kV Wire to Ground	±1kV Wire to Wire Not Applicable	The network power supply shall be equipped with the quality used in the classic commercial or medical environment.
Temporary Reduction, Temporary Interruption and Voltage Change of the Input Line of the Power Supply GB/T 17626.11	<5% U_T, Continuous for 0.5 Cycle (In U_T, >95% Temporary Reduction) 40% U_T, Continuous for 5 Cycle (In U_T, >60% Temporary Reduction) 70% U_T, Continuous for 25 Cycle (In U_T, >30% Temporary Reduction) <5% U_T, Continuous for 5s (In U_T, > 95% Temporary Reduction)	<5% U_T, Continuous for 0.5 Cycle (In U_T, >95% Temporary Reduction) 40% U_T, Continuous for 5 Cycle (In U_T, >60% Temporary Reduction) 70% U_T, Continuous for 25 Cycle (In U_T, >30% Temporary Reduction) <5% U_T, Continuous for 5s (In U_T, > 95% Temporary Reduction)	The network power supply shall be equipped with the quality used in the classic commercial or medical environment. If the continuous operation needs to be conducted by the user for the use of the Video Laryngoscopes during the interruption period of the power supply, it is recommended to adopt the uninterruptible power supply or battery for the Video Laryngoscopes for the power supply.
Power Frequency Magnetic Field (50Hz/60Hz) GB/T 17626.8)	3A/m	3A/m	The power frequency magnetic field shall be equipped with the level characteristics of the power frequency magnetic field of the classic locations of the classic commercial or hospital environment.
Note: U_T refers to the voltage of the AC network before the implementation and test of voltage.			

Guideline and Statement of the Manufacturer – Electromagnetic Immunity (Table 2.0)

Guideline and Statement of the Manufacturer – Electromagnetic Immunity			
The Video Laryngoscopes is estimated to be used in the following stipulated electromagnetic environment, and the purchasers or users shall guarantee that it is used in the electromagnetic environment.			
Immunity Test	IEC 60601 Test Level	Confo-rmity Level	Electromagnetic Environment – Guideline
Radio-frequency Conduction GB/T 17626.6 Radio-frequency Radiation GB/T 17626.3	3V (Valid Value) 150 kHz-80kHz 3V/m 80MHz-2.5GHz	3V (Valid Value) 3V/m	The portable and mobile radio-frequency communication equipment shall not get close to any part of the Video Laryngoscopes beyond the recommended isolation distance, including the cables, and the distance shall be calculated in accordance with the format corresponding to the frequency of transmitter. ***Recommended Isolation Distance:*** $d=1.2\sqrt{p}$ $d=1.2\sqrt{p}$ 80MHz-800MHz $d=2.3\sqrt{p}$ 800MHz-2.5GHz ***Where:*** ***p***-The maximum rated output power of the transmitter according to the transmitter manufacture, and the unit is: Watt (W); ***d*** -Recommended isolation distance, unit: Meter (m) Field strength from fixed RF transmitters, as determined by an electromagnetic stie survey, [a] should be less than the compliance level in each frequency range [b] . The disturbance may happen to the surrounding of the equipment with the following sign.

Note 1: If the frequency point is 80MHz and 800MHz, the format of higher frequency section shall be adopted.

Note 2: Those guidelines may not be suitable for all situations, and the electromagnetic transmission is affected by the absorption and emission of the buildings, articles, and bodies.

a Fixed transmitter, like: Radio (cellular/cordless) telephone and ground mobile wireless base station, amateur radio, amplitude modulation, FM radio, television broadcasting, etc., and their field intensities cannot be accurately known in theory. The survey of the electromagnetic field shall be considered for evaluating the electromagnetic environment of the fixed radio-frequency transmitter. If the measured filed intensity of the places where the Video laryngoscopes locates higher than the above applicable radio frequency meets the level, the Video Laryngoscopes shall be observed to verify its normal operation. If the abnormal performance cannot be observed, the supplementary measures may be needed, and the direction or position of the Video Laryngoscopes is adjusted again.

b The field intensity shall be lower than 3V/m, within the whole frequency scope of 150kHz-80MHz.

Recommendable Isolation Distance of the Portable and Mobile Radio-frequency Communication Equipment and Video Laryngoscopes (Table 2.1)

Recommendable Isolation Distance of the Portable and Mobile Radio-frequency Communication Equipment and Video Laryngoscopes .
It is estimated that the Video Laryngoscopes will be used in the electromagnetic environment where the radio-frequency radiation disturbance is controlled. In accordance with the maximum rated output power of the communication equipment, the purchasers or users can maintain the minimum distance of the portable and mobile radio-frequency communication equipment (transmitter) and Video Laryngoscopes through the following recommendation to prevent the electromagnetic interference.

Maximum Rated Output Power of the Transmitter W	Isolation Distance Corresponding to Different Frequencies of the Transmitter / m		
	150kHz-80MHz $d= 1.2\sqrt{P}$	80MHz-800MHz $d= 1.2\sqrt{P}$	800MHz-2.5GHz $d= 2.3\sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23

For the maximum rated output power of the transmitter not listed in the above table, the recommended distance d takes the meter (m) as the unit, and can be confirmed by the format in the corresponding frequency column of the transmitter, and here P is the maximum rated output power of the transmitter provided by the transmitter manufacturer, and the unit is Watt (W).

Note 1: If the frequency point is 80MHz and 800MHz, the format of higher frequency section shall be adopted.

Note 2: Those guidelines may not be suitable for all situations, and the electromagnetic transmission is affected by the absorption and emission of the buildings, articles, and bodies.